Arnab Chakraborty

Full Stack Developer

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Education

Gauhati University, Bachelor of Technology (B.Tech.) in Information Technology — Aug 2023 - June 2027 (Expected)

Relevant Coursework: Computer Science fundamentals, Data Structures, OOP, DBMS, Operating Systems

Experience

Full Stack Developer Intern, FRINT — Mar 2026 - Apr 2026

- Built an **AI proctoring system** with React + Vite frontend and Express/Socket.io gateway.
- Developed **Python FastAPI microservices** for AI/ML using Whisper STT, MediaPipe, OpenCV, MongoDB, Redis, Groq, and Gemini APIs.
- Technologies: React.js, TypeScript, Python, FastAPI, Node.js, Express, Socket.io, MongoDB, Redis, WebRTC, Groq, Gemini, Whisper, MediaPipe, OpenCV

Research Intern, National Institute of Technology, Silchar — Jun 2026 - Jul 2026

- Built a **multimodal deep learning pipeline** under Dr. Ram Kumar Karsh on the RASTA retinal OCTA dataset, fusing three retinal layers, clinical data, and U-Net++ segmentation masks through SimCLR-pretrained ConvNeXt encoders and cross-layer attention.
- Reached **0.80 patient-level accuracy** and **0.898 macro ROC-AUC** under leakage-safe patient-grouped 5-fold cross-validation, with temperature calibration, MC-dropout uncertainty, and Grad-CAM/SHAP explainability.
- Technologies: Python, PyTorch, ConvNeXt, U-Net++, SimCLR, OpenCV, scikit-learn, SHAP

Open-Source Contributor, MetaBrainz Picard — Nov 2024 - Dec 2024

- Implemented **side-by-side old and new file paths** in the rename preview. (PR #2563)
- Built a custom title-casing function fixing **100+ capitalization issues** and improving metadata consistency; added Pytest unit tests increasing coverage. (PR #2548)
- Technologies: Git, GitHub, Python, PyQT, Pytest

Projects

System - 2026 — system.zytact.com

- Built a personal-progression web app where external AI agents submit proposals through an authenticated MCP server, but **only the human can confirm** anything that alters the permanent record.
- Isolated all scoring in a **standalone engine** that derives progression from pinned rules, so neither the agent surface nor the UI can award points directly.
- Technologies: TypeScript, Next.js, Convex, Clerk, MCP, Vitest

InvenTrack - 2025 — inventrack.zytact.com

- Built a **multi-tenant inventory platform** with per-tenant data isolation, automated reorder triggers, and dynamic pricing rules in a single PostgreSQL schema.
- Integrated **Gemini LLM** to generate demand forecasts and restock recommendations from live inventory data, requiring careful prompt design to produce reliable, structured output.
- Technologies: Next.js, Clerk, TypeScript, Tailwind CSS, Shadcn UI, TanStack Query, Drizzle ORM, Gemini API

Compress - 2026 — compress.zytact.com

- Built a **privacy-first image compressor** using Rust/WASM with Lanczos resampling and target-size binary search; **100% client-side**, no server uploads or tracking.
- Technologies: Rust, WebAssembly, TypeScript, TanStack Start

Technical Skills

Programming Languages: Python, C, JavaScript, TypeScript, HTML, CSS

AI / Machine Learning: PyTorch, scikit-learn, NumPy, Pandas, Computer Vision, Hugging Face Transformers, OpenAI API, Gemini API, Large Language Models (LLMs), Prompt Engineering, Generative AI

Front End Web: React.js, Next.js, Tailwind CSS, TanStack Query, Zustand, React Konva, Web design

Back End: Node.js, Express.js, Python, RESTful APIs, tRPC, Auth.js, Better Auth, Clerk

Database: MongoDB, PostgreSQL, MySQL

Tools: Git, GitHub Actions, Linux, Vim/NeoVim, Jest, Vitest, Pytest, ESLint, Prettier